CREDIT RISK ANALYTICS

Loan Default Prediction

using Machine Learning & Explainable AI

End-to-End Credit Risk Modelling Project

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Dataset: HMEQ (Home Equity) | 5,960 Records | 13 Variables

Python 3.12 | Scikit-learn | XGBoost | SHAP

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1. Project Overview & Business Objective

A major portion of retail bank revenue is derived from interest income on home equity loans. While these loans generate significant profits, they simultaneously expose financial institutions to **credit risk** — the risk that a borrower will fail to repay the loan as agreed. Each undetected defaulter represents a direct financial loss (Non-Performing Asset / NPA) that erodes bank profitability.

The consumer credit department requires an **empirically derived, statistically sound, and interpretable** credit scoring model — in compliance with the Equal Credit Opportunity Act — that can automate loan risk assessment while providing justifiable, explainable predictions for any adverse decision (rejection).

Primary Objective

Build a supervised binary classification model that predicts whether a home equity loan applicant is likely to **default (BAD=1)** or **repay (BAD=0)**, with the primary goal of minimizing the number of defaulters incorrectly classified as non-defaulters.

Key Business Questions

- Can the bank identify applicants who are likely to miss payments or completely default before the loan is approved?
- Which applicant characteristics are the strongest indicators of default risk?
- How can the bank reduce financial losses (NPAs) caused by high-risk borrowers?
- How do we ensure that automated credit decisions are transparent and explainable?

Evaluation Strategy

Since the dataset is **imbalanced (~20% defaulters)**, simple accuracy is a misleading metric. The primary metric is **Recall** (sensitivity), which measures the model's ability to correctly identify all actual defaulters. A missed defaulter (False Negative) is far more costly than a false alarm (False Positive).

Metric	Why It Matters for Credit Risk
Recall (Primary)	Captures maximum defaulters — minimizes missed bad loans
Precision	Measures how many flagged applicants are genuinely risky
F1-Score	Balances Precision and Recall — useful with class imbalance
ROC-AUC	Overall discrimination ability across all classification thresholds
PR-AUC	Especially informative when positive class is rare (defaulters ~20%)

2. Dataset Description

The **Home Equity (HMEQ) Dataset** contains baseline and loan performance information for **5,960 recent home equity loans** issued by a retail bank. Each record captures applicant demographics, financial history, loan characteristics, and credit behavior.

Variable	Type	Description
BAD	Target (Binary)	1 = Defaulted / Severely Delinquent; 0 = Loan Repaid
LOAN	Numerical	Approved loan amount (INR/USD)
MORTDUE	Numerical	Outstanding balance on existing mortgage
VALUE	Numerical	Current appraised property value (collateral)
REASON	Categorical	Loan purpose: HomeImp (home improvement) or DebtCon (debt consolidation)
JOB	Categorical	Occupation category: Mgr, Office, Other, ProfExe, Sales, Self
YOJ	Numerical	Years at current employer
DEROG	Numerical	Number of major derogatory reports (serious delinquencies)
DELINQ	Numerical	Number of delinquent credit lines (30–60 days past due)
CLAGE	Numerical	Age of oldest credit line in months (credit history length)
NINQ	Numerical	Number of recent credit inquiries
CLNO	Numerical	Total number of existing credit lines
DEBTINC	Numerical	Debt-to-Income ratio: monthly debt / gross monthly income

Class Distribution

Class	Label	Count	Percentage
Non-Default	BAD = 0	4,771	80.05%
Default	BAD = 1	1,189	19.95%
Total	—	5,960	100.00%

The dataset is **moderately imbalanced**. The default rate (~20%) is significant enough to require careful model evaluation but not so extreme as to require mandatory resampling techniques, though class-weight adjustments are considered.

3. Data Overview & Cleaning

3.1 Dataset Shape & Structure

Initial dataset dimensions: **5,960 rows × 13 columns**. No duplicate records were found.

Metric	Detail
Total Records	5,960
Duplicate Records	0 (none found)
Numerical Columns	10 (LOAN, MORTDUE, VALUE, YOJ, DEROG, DELINQ, CLAGE, NINQ, CLNO, DEBTINC)
Categorical Columns	3 (BAD, REASON, JOB)
Target Variable	BAD (binary: 0 / 1)

3.2 Missing Value Analysis

Variable	Missing Count	Missing %	Variable Type
DEBTINC	1,267	21.26%	Numerical
DEROG	708	11.88%	Numerical
DELINQ	580	9.73%	Numerical
MORTDUE	518	8.69%	Numerical
YOJ	515	8.64%	Numerical
NINQ	510	8.56%	Numerical
CLAGE	308	5.17%	Numerical
JOB	279	4.68%	Categorical
REASON	252	4.23%	Categorical
CLNO	222	3.72%	Numerical
VALUE	112	1.88%	Numerical

3.3 Data Type Conversion

- BAD converted from int64 → category (binary target variable).
- REASON and JOB converted from object → category (nominal categorical).
- 10 numerical variables retained as float64 / int64.

3.4 Initial Observations

- No duplicate records found — all 5,960 observations are unique.
- DEBTINC has the highest missingness (21.26%) and is also one of the most important predictors of default.
- Several variables contain extreme right skewness, typical of financial datasets.
- Two categorical variables (REASON, JOB) require encoding before model development.
- The moderate class imbalance (~20% defaulters) necessitates careful metric selection — Recall prioritized over Accuracy.

4. Exploratory Data Analysis (EDA)

EDA was performed using a structured Univariate → Bivariate → Multivariate → Correlation framework. A total of **15 visualizations** were created to uncover patterns, relationships, and business insights embedded in the data.

4.1 Univariate Analysis

Chart 1 – Target Variable Distribution (Count Plot + Pie Chart)

80.05% of applicants successfully repaid their loans; 19.95% defaulted. The dataset is moderately imbalanced, confirming that Recall and F1-Score should be the primary evaluation metrics rather than raw accuracy.

Chart 2 – Numerical Variable Histograms

All 10 numerical variables were plotted. Most variables exhibit significant positive skewness — LOAN, MORTDUE, VALUE, and DEBTINC all have long right tails, indicating a subset of customers with unusually high financial obligations. This skewness also explains the presence of outliers.

Chart 3 – KDE (Kernel Density) Plots

KDE plots confirmed the right-skewed distribution across financial variables. DEROG and DELINQ show a concentration at zero, indicating that most customers have clean credit histories, with a smaller segment carrying multiple derogatory marks.

Chart 4 – Boxplots (Outlier Visualization)

Boxplots revealed extreme outliers across most financial variables. These values represent genuine high-risk or high-value customers and are expected in banking datasets. They were capped rather than removed during outlier treatment.

Chart 5 – Categorical Variable Count Plots (REASON & JOB)

Debt consolidation (DebtCon) is the dominant loan purpose, outnumbering home improvement (HomeImp) requests. Among occupations, 'Other' is the most common category, followed by 'ProfExe' and 'Mgr'. These categories show differential default rates.

4.2 Bivariate Analysis

Chart 6 – Job Type vs Loan Default (Count Plot)

Default rates vary across occupational categories. Self-employed individuals and those in sales roles appear to exhibit relatively higher default rates, possibly due to income variability. Professional executive (ProfExe) and managerial roles show lower relative default proportions.

Chart 7 – Loan Purpose vs Default Rate (Grouped Bar Chart)

Debt consolidation borrowers show a marginally higher default rate compared to home improvement borrowers, suggesting that applicants already managing multiple debts may face greater repayment challenges.

Chart 8 – Debt-to-Income Ratio vs Default (Violin Plot)

Defaulted customers exhibit a noticeably higher DEBTINC distribution compared to non-defaulters. This confirms that debt-to-income ratio is one of the strongest predictors of credit risk — borrowers spending a larger share of income on debt are significantly more likely to default.

Chart 9 – Loan Amount vs Default (Box Plot)

Defaulters tend to have slightly higher approved loan amounts. Larger loan obligations increase repayment burden, particularly when combined with high existing debt levels.

Chart 10 – Loan Amount vs Property Value (Scatter Plot)

Customers requesting large loans relative to low property values exhibit higher default risk. This ratio (Loan-to-Value) is a critical credit risk indicator — lower collateral value reduces the bank's recovery in case of default.

4.3 Multivariate Analysis

Chart 11 – Pair Plot (LOAN, VALUE, MORTDUE, DEBTINC by BAD)

Pair plots reveal noticeable separation between default and non-default clusters across financial variables. DEBTINC and LOAN show the clearest separation, confirming their predictive power. The interaction between high MORTDUE and low VALUE also characterizes defaulters.

Chart 12 – Mortgage Due vs Property Value (Grouped Scatter)

Applicants with high outstanding mortgage balances and relatively lower property values cluster in the defaulter category. This confirms the importance of collateral coverage in credit risk.

Chart 13 – Random Forest Feature Importance

The Random Forest baseline model's feature importance rankings confirm the EDA findings: DEBTINC, CLAGE, VALUE, MORTDUE, and LOAN emerge as the top contributors to default risk, consistent with credit risk domain knowledge.

4.4 Correlation Analysis (Heatmap)

A correlation heatmap of all 10 numerical variables was generated. Key findings:

- **MORTDUE and VALUE** are strongly positively correlated (~0.87), indicating that properties with higher values tend to have larger outstanding mortgages.
- **LOAN and VALUE** show moderate positive correlation — larger loans tend to be backed by higher-value properties.
- **DEBTINC** exhibits limited correlation with other variables, underscoring its independent predictive value.
- **DEROG and DELINQ** show moderate positive correlation, both capturing credit deterioration.
- No severe multicollinearity exists that would undermine model stability.

4.5 Key Business Insights from EDA

Risk Factor	Business Implication
Debt-to-Income Ratio (DEBTINC)	Strongest single predictor of default. High DEBTINC → significantly elevated default probability.
Credit Delinquencies (DELINQ)	Number of delinquent credit lines strongly predicts default — past behavior predicts future behavior.
Derogatory Reports (DEROG)	Major negative credit events (DEROG) are powerful early warning signals.
Property Value (VALUE)	Higher collateral value reduces both default probability and potential recovery losses.
Credit Inquiry Frequency (NINQ)	Frequent recent credit inquiries signal financial distress — borrowers seeking multiple credit lines.
Employment Stability (YOJ)	Shorter tenure at current employer correlates with higher default risk.
Loan Purpose	Debt consolidation borrowers already managing high debt levels may face compounded repayment challenges.

5. Data Preprocessing Pipeline

A comprehensive preprocessing pipeline was implemented to prepare the raw HMEQ dataset for machine learning model development. The pipeline consists of six sequential steps.

Step 1 – Missing Value Treatment

Method: Median Imputation (Numerical) + Mode Imputation (Categorical)

Rationale: Financial variables are heavily right-skewed with outliers. The median is more robust than the mean in skewed distributions and is standard practice in credit risk preprocessing. Mode imputation preserves the most common category for REASON and JOB without introducing artificial values.

- **After treatment:** 0 missing values across all 13 columns

Step 2 – Outlier Treatment (Winsorization / IQR Capping)

Method: IQR-Based Capping

Rationale: Outliers in banking data are not errors — they represent genuine high-value or high-risk customers. Removing them would discard valuable credit risk signals. IQR capping clips extreme values to [Q1 - 1.5×IQR, Q3 + 1.5×IQR] bounds, reducing their disproportionate influence on model training while preserving the observations.

- **Method:** cap_outliers() function applied to all 10 numerical columns
- **Approach:** Values above upper bound → capped at upper; values below lower → capped at lower

Step 3 – Categorical Encoding (One-Hot Encoding)

Method: pd.get_dummies() with drop_first=True

Rationale: Machine learning algorithms require numerical inputs. One-Hot Encoding is appropriate for REASON and JOB because no ordinal relationship exists between categories. drop_first=True avoids the Dummy Variable Trap (perfect multicollinearity).

- **REASON:** 2 categories → 1 dummy (REASON_HomeImp)
- **JOB:** 6 categories → 5 dummies (JOB_Office, JOB_Other, JOB_ProfExe, JOB_Sales, JOB_Self)
- **Final shape:** (5,960 × 17) after encoding

Step 4 – Feature/Target Separation

Method: X (16 features) and y (BAD) separated

Rationale: The target variable BAD is separated from the feature matrix. The 16 independent variables comprise 10 original numerical features plus 6 one-hot encoded dummy variables.

- **X shape:** (5,960 × 16)
- **y shape:** (5,960,)

Step 5 – Train-Test Split

Method: 80% Train / 20% Test | Stratified Sampling | random_state=42

Rationale: Stratified splitting ensures that the proportion of defaulters (BAD=1) is preserved identically in both training and testing sets. This prevents the testing set from being unrepresentative of the actual default rate.

- **Training set:** 4,768 records (≈ 80%)
- **Test set:** 1,192 records (≈ 20%)
- **Default rate preserved:** ~20% in both splits

Step 6 – Feature Scaling (StandardScaler)

Method: Standardization: Mean=0, Std=1

Rationale: Feature scaling is critical for Logistic Regression, which uses gradient-based optimization. Without scaling, variables with larger numeric ranges dominate the objective function. Tree-based models (Random Forest, XGBoost) are scale-invariant and do not strictly require scaling, but scaled data was prepared for consistent comparison.

- **Scaled on:** Training data (fit_transform)
- **Applied to:** Test data (transform only — prevents data leakage)

6. Baseline Machine Learning Models

Three supervised classification algorithms were trained on the preprocessed HMEQ dataset. All three models were evaluated using a common evaluation function that computes Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, ROC Curve, and Precision-Recall Curve.

6.1 Logistic Regression

Why Logistic Regression? Logistic Regression is the industry-standard baseline for credit scoring. It is highly interpretable, produces well-calibrated probability estimates, and complies with regulatory requirements for explainable credit decisions (Equal Credit Opportunity Act). Configuration: max_iter=1000, random_state=42.

Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
80.45%	55.81%	10.08%	17.08%	67.85%	36.64%

Class	Precision	Recall	F1-Score	Support
Non-Default (0)	0.81	0.98	0.89	954
Default (1)	0.56	0.10	0.17	238

Analysis: Logistic Regression achieves 80% accuracy, but the Recall for defaulters is catastrophically low at 10.08% — it correctly identifies only ~10 out of every 100 defaulters. This makes it unsuitable as a standalone credit risk model without significant improvement. The low PR-AUC (36.64%) confirms poor performance on the minority default class.

6.2 Random Forest

Why Random Forest? Random Forest is an ensemble of 100 decision trees (default) that handles nonlinear feature interactions robustly and is resistant to outliers. It does not require feature scaling. Configuration: n_estimators=100 (default), random_state=42.

Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
88.93%	76.50%	64.29%	69.86%	95.16%	83.02%

Class	Precision	Recall	F1-Score	Support
Non-Default (0)	0.91	0.95	0.93	954
Default (1)	0.77	0.64	0.70	238

Analysis: Random Forest delivers a dramatic improvement over Logistic Regression. Recall for defaulters jumps from 10.08% to 64.29%, and ROC-AUC reaches 95.16%, indicating excellent discriminatory power. The PR-AUC of 83.02% confirms strong performance even on the imbalanced minority class. This model captures nonlinear credit risk patterns effectively.

6.3 XGBoost

Why XGBoost? Extreme Gradient Boosting builds trees sequentially, with each tree correcting the errors of the previous one. It includes built-in regularization (gamma, lambda) and is widely used in industry credit risk models. Configuration: eval_metric='logloss', random_state=42.

Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
89.43%	80.11%	62.61%	70.28%	94.28%	83.70%

Class	Precision	Recall	F1-Score	Support
Non-Default (0)	0.91	0.96	0.94	954

Default (1)	0.80	0.63	0.70	238

Analysis: XGBoost achieves the highest overall accuracy (89.43%) and precision (80.11%) among baseline models. Its Recall (62.61%) is slightly below Random Forest, but its higher precision means fewer false alarms. PR-AUC (83.70%) is the highest among baselines. Hyperparameter tuning is expected to improve recall further.

6.4 Baseline Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Logistic Regression	80.45%	55.81%	10.08%	17.08%	67.85%	36.64%
Random Forest	88.93%	76.50%	64.29%	69.86%	95.16%	83.02%
XGBoost	89.43%	80.11%	62.61%	70.28%	94.28%	83.70%

Baseline Ranking (by Recall): Random Forest (64.29%) > XGBoost (62.61%) >> Logistic Regression (10.08%). Both ensemble models significantly outperform Logistic Regression on the primary metric, with Random Forest showing the best recall at baseline.

Threshold Analysis: At a reduced decision threshold of 0.35 (instead of default 0.5), XGBoost achieves Recall=73%, Precision=76%, F1=75%, and Accuracy=90%, demonstrating that threshold optimization can further improve model utility without retraining.

7. Hyperparameter Tuning

All three models were systematically tuned to find the optimal hyperparameter configurations. Tuning was performed using cross-validation, with **Recall as the primary scoring metric** to prioritize correct identification of defaulters.

7.1 Logistic Regression — GridSearchCV

GridSearchCV exhaustively evaluates every combination in the parameter grid. Logistic Regression has few hyperparameters, making exhaustive search computationally feasible.

Hyperparameter	Search Grid	Best Value
C (Regularization)	[0.01, 0.1, 1, 10, 100]	From CV
penalty	['l2']	l2
solver	['lbfgs', 'liblinear']	From CV

Search Method: GridSearchCV | 5-fold CV | Scoring: Recall | n_jobs=-1

Accuracy	Precision	Recall	F1-Score	ROC-AUC
80.45%	**55.81%**	**10.08%**	**17.08%**	**67.85%**

Result: Hyperparameter tuning did not improve Logistic Regression's performance. The default configuration was already close to optimal, confirming that the linear model's limitation stems from the nonlinear nature of credit risk, not parameter settings.

7.2 Random Forest — RandomizedSearchCV

Random Forest has many hyperparameters. RandomizedSearchCV samples 20 combinations from the parameter distribution, balancing exploration efficiency with computational cost.

Hyperparameter	Search Grid
n_estimators	[100, 200, 300, 500]
max_depth	[5, 10, 15, 20, None]
min_samples_split	[2, 5, 10]
min_samples_leaf	[1, 2, 4]
max_features	['sqrt', 'log2']

Search Method: RandomizedSearchCV | n_iter=20 | 5-fold CV | Scoring: Recall | random_state=42

Accuracy	Precision	Recall	F1-Score	ROC-AUC
89.01%	**77.44%**	**63.45%**	**69.75%**	**94.66%**

Result: Tuned Random Forest shows incremental improvement in Accuracy (+0.08%) and Precision (+0.94%), but Recall decreased marginally (-0.84%). The model remains highly competitive with ROC-AUC of 94.66%.

7.3 XGBoost — RandomizedSearchCV

XGBoost contains numerous hyperparameters controlling tree structure, regularization, and ensemble behavior. RandomizedSearchCV samples 20 combinations efficiently.

Hyperparameter	Search Grid
n_estimators	[100, 200, 300]
learning_rate	[0.01, 0.05, 0.1]

max_depth	[3, 5, 7]
subsample	[0.7, 0.8, 1.0]
colsample_bytree	[0.7, 0.8, 1.0]
gamma	[0, 0.1, 0.3]

Search Method: RandomizedSearchCV | n_iter=20 | 5-fold CV | Scoring: Recall | random_state=42

Accuracy	Precision	Recall	F1-Score	ROC-AUC
90.94%	**84.21%**	**67.23%**	**74.77%**	**95.26%**

Result: Tuned XGBoost shows the most significant improvement. Accuracy increased to 90.94%, Precision jumped to 84.21% (+4.1%), Recall improved to 67.23% (+4.6%), F1-Score rose to 74.77% (+4.5%), and ROC-AUC reached 95.26%. This is the strongest performer across all metrics after tuning.

8. Final Model Selection & Comparison

8.1 Complete Baseline vs Tuned Comparison

Model	Type	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	Baseline	80.45%	55.81%	10.08%	17.08%	67.85%
Tuned LR	Tuned	80.45%	55.81%	10.08%	17.08%	67.85%
Random Forest	Baseline	88.93%	76.50%	64.29%	69.86%	95.16%
Tuned Random Forest	Tuned	89.01%	77.44%	63.45%	69.75%	94.66%
XGBoost	Baseline	89.43%	80.11%	62.61%	70.28%	94.28%
Tuned XGBoost	Tuned	90.94%	84.21%	67.23%	74.77%	95.26%

8.2 Final Model: Tuned XGBoost

FINAL MODEL: Tuned XGBoost Accuracy: 90.94% | Precision: 84.21% | Recall: 67.23% | F1-Score: 74.77% | ROC-AUC: 95.26%

Class	Precision	Recall	F1-Score	Support
Non-Default (0)	0.92	0.97	0.94	954
Default (1)	0.84	0.67	0.75	238

8.3 Why Tuned XGBoost?

- **Highest Recall (67.23%):** Correctly identifies the most defaulters among all evaluated models — directly addressing the primary business objective of minimizing undetected bad loans.
- **Highest Precision (84.21%):** When the model flags an applicant as a defaulter, it is correct 84.21% of the time, minimizing unnecessary loan rejections for creditworthy applicants.
- **Best F1-Score (74.77%):** The strongest balance between Precision and Recall across all models.
- **Best ROC-AUC (95.26%):** Highest discriminatory power across all classification thresholds, confirming the model's reliability for rank-ordering applicants by risk.
- **Gradient Boosting Architecture:** Sequential error correction makes XGBoost particularly effective at capturing the complex nonlinear patterns in credit risk data.
- **Regularization:** Built-in L1/L2 regularization and tree pruning (gamma) prevent overfitting, making the tuned model generalizable to new loan applicants.

9. Explainable AI using SHAP

The final Tuned XGBoost model is a high-accuracy 'black-box' whose internal decision logic is not directly interpretable. To ensure regulatory compliance, transparency, and stakeholder trust, **SHAP (SHapley Additive exPlanations)** was applied to explain model predictions at both global and individual customer levels.

SHAP is grounded in cooperative game theory (Shapley values) and provides theoretically consistent, model-agnostic feature attributions. A **TreeExplainer** was used — the most computationally efficient SHAP explainer for tree-based models.

9.1 Why SHAP?

- SHAP provides both **global explanations** (which features matter most overall) and **local explanations** (why was THIS specific customer predicted as a defaulter).
- Unlike simple feature importance, SHAP quantifies both the **direction** (risk-increasing or risk-decreasing) and **magnitude** of each feature's contribution.
- SHAP supports **regulatory compliance** with the Equal Credit Opportunity Act — the bank can provide feature-level justifications for adverse credit decisions.
- SHAP explanations help credit officers **override or validate** model predictions using business judgment.

9.2 Global Feature Importance (SHAP Ranking)

Rank	Feature	SHAP Importance	Direction of Risk
1	DEBTINC	1.7221	High DEBTINC → Higher default risk
2	CLAGE	0.7166	Low CLAGE (shorter credit history) → Higher risk
3	VALUE	0.4995	Low property value → Higher default risk
4	MORTDUE	0.4861	High outstanding mortgage → Higher risk
5	LOAN	0.4678	Higher loan amount → Elevated risk
6	YOJ	0.4137	Fewer years at employer → Higher risk
7	CLNO	0.3966	Fewer credit lines (thin file) → Higher risk
8	NINQ	0.2969	More recent inquiries → Higher default risk
9	JOB_Office	0.2383	Office job category — moderate protective effect
10	REASON_HomeImp	0.1400	Home improvement purpose — lower risk vs DebtCon

9.3 SHAP Visualization Outputs

Beeswarm Plot (SHAP Summary)

Displays the distribution of SHAP values across all test observations for each top-15 feature. Each dot represents one customer. Red dots indicate higher feature values; blue indicates lower values. DEBTINC appears at the top with the widest spread, confirming its dominant influence on default predictions.

SHAP Bar Plot (Global Feature Importance)

Ranks features by their mean absolute SHAP value. DEBTINC, CLAGE, VALUE, MORTDUE, and LOAN are the top five most influential variables globally, consistent with credit risk domain knowledge.

Waterfall Plot (Local Explanation)

Explains why the model predicted a specific customer's default probability. Each feature's contribution is shown as a positive (red) or negative (blue) bar relative to the baseline expected value. The final output is the customer's personalized predicted risk score.

Force Plot (Individual Prediction)

Visualizes the push-and-pull of features on a single prediction. Features in red push the prediction toward higher default probability; features in blue push it toward lower risk. Enables credit officers to explain rejections at the applicant level.

SHAP Dependence Plot (DEBTINC)

Shows how changes in DEBTINC values influence SHAP impact on predictions. A clear non-linear pattern emerges — beyond a critical DEBTINC threshold, default probability increases sharply, revealing the non-linear credit risk threshold effect.

Multi-Customer Waterfall Plots

Waterfall plots for three individual customers reveal that the same features can have opposite effects depending on the customer's profile — demonstrating the model's individualized (not one-size-fits-all) risk assessment capability.

9.4 SHAP Business Insights

- A borrower with DEBTINC above ~40% combined with low CLAGE and multiple recent credit inquiries carries the highest predicted default risk and should receive enhanced due diligence.
- Property value (VALUE) acts as a buffer — high collateral partially offsets other risk factors by providing recovery options in case of default.
- Employment tenure (YOJ) serves as an income stability proxy — applicants with fewer than 2 years at their current employer show elevated risk scores.
- Applicants requesting loans for debt consolidation with high existing DEBTINC represent the highest-risk segment in the portfolio.
- SHAP local explanations can be directly embedded into rejection letters, providing personalized, legally defensible adverse action notices.

10. Business Recommendations

Based on the combined findings from EDA, predictive modeling, and SHAP explainability, the following actionable recommendations are proposed for the bank's credit risk department.

- **1. Deploy DEBTINC Screening as a Primary Credit Gate**

Given DEBTINC's dominant role (SHAP Importance: 1.72), the bank should implement an automated pre-screening rule: applicants with DEBTINC above a defined threshold (e.g., 40%) should be automatically escalated to manual underwriting review before loan consideration.

- **2. Multi-Factor Risk Scoring System**

Replace single-metric screening with a composite credit score integrating the top SHAP features: DEBTINC + CLAGE + VALUE + DELINQ + DEROG + NINQ. This holistic approach reduces both false approvals (missed defaulters) and false rejections (creditworthy applicants).

- **3. Integrate SHAP Explanations into Loan Officer Workflows**

Provide loan officers with applicant-specific SHAP waterfall plots alongside each model prediction. This enables human oversight, supports explainable adverse action notices, and ensures compliance with the Equal Credit Opportunity Act.

- **4. Optimize Decision Threshold (0.35 instead of 0.50)**

Lowering the classification threshold to 0.35 increases Recall from 67% to ~73% with minimal precision loss. The bank should conduct a cost-benefit analysis to identify the optimal threshold that balances the cost of missed defaulters vs. rejected creditworthy applicants.

- **5. Risk-Based Loan Pricing**

The model's predicted default probability can be used to apply risk-adjusted interest rates. High-risk borrowers (BAD probability > 0.4) may be offered loans with higher rates or stricter collateral requirements, rather than outright rejection.

- **6. Flag Debt Consolidation Applicants with High DEBTINC**

Applicants combining high DEBTINC with a debt consolidation purpose represent a compounded risk profile — already over-leveraged and seeking additional credit. This segment warrants enhanced scrutiny.

- **7. Continuous Model Retraining & Monitoring**

Deploy the model in a production environment with monitoring for concept drift. Schedule quarterly retraining as new loan performance data accumulates, ensuring the model adapts to evolving borrower behavior and macroeconomic conditions.

- **8. Address Class Imbalance in Future Iterations**

Future model versions should explore SMOTE oversampling, class_weight='balanced' parameters, and cost-sensitive learning to further improve recall for the minority defaulter class without sacrificing precision.

11. Conclusion & Future Scope

This project delivered a complete, production-quality end-to-end credit risk modelling solution for home equity loan default prediction. Starting from raw HMEQ data, the project addressed every stage of the data science lifecycle:

- Comprehensive data understanding and quality assessment with structured missing value analysis.
- 15-chart EDA framework revealing key credit risk drivers and segment-level default patterns.
- Robust preprocessing pipeline: median/mode imputation, IQR Winsorization, One-Hot Encoding, stratified train-test split, and StandardScaler.
- Development and evaluation of three baseline classifiers with Confusion Matrix, ROC Curves, Precision-Recall Curves, and PR-AUC analysis.
- Hyperparameter optimization using GridSearchCV (Logistic Regression) and RandomizedSearchCV (Random Forest, XGBoost) with Recall as the primary optimization metric.
- Selection of Tuned XGBoost as the final model: Accuracy=90.94%, Precision=84.21%, Recall=67.23%, F1=74.77%, ROC-AUC=95.26%.
- Explainable AI using SHAP TreeExplainer: global feature importance, beeswarm plots, bar plots, waterfall plots, force plots, and dependence plots.
- 8 actionable business recommendations for responsible, equitable credit risk deployment.

Key Takeaway

The Tuned XGBoost model achieves a 95.26% ROC-AUC and correctly identifies 67.23% of defaulters with 84.21% precision — enabling the bank to significantly reduce NPA losses while maintaining fair, explainable, and regulatory-compliant lending decisions.

Future Scope

- **SMOTE & Class-Weight Techniques:** Apply Synthetic Minority Oversampling (SMOTE) or class_weight='balanced' to further boost Recall for the underrepresented defaulter class.
- **Probability Calibration:** Apply Platt Scaling or Isotonic Regression to ensure XGBoost's probability outputs are well-calibrated for risk-based pricing and threshold optimization.
- **Cost-Sensitive Learning:** Integrate asymmetric misclassification costs directly into the loss function — penalizing missed defaulters (False Negatives) more heavily than false alarms.
- **Ensemble Stacking:** Combine predictions from Logistic Regression, Random Forest, and XGBoost using a meta-learner to potentially improve overall performance.
- **Real-Time Credit Scoring API:** Deploy the model as a REST API (FastAPI/Flask) for real-time loan application scoring, returning both the binary prediction and an applicant-specific SHAP explanation.
- **Survival Analysis:** Model the time-to-default using survival analysis techniques (Cox PH, Kaplan-Meier) to predict not just whether default occurs, but when.
- **Deep Learning Models:** Explore neural network architectures (TabNet, AutoML) for potentially higher accuracy on the structured HMEQ feature set.